APOLLO 13

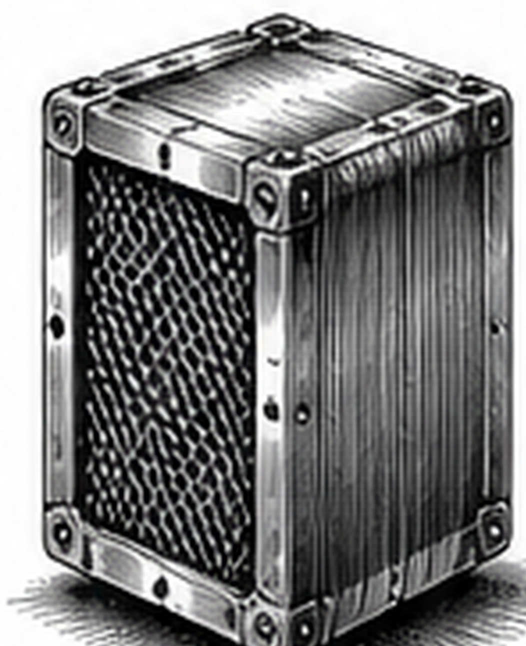
SPARKWORKS

Think through anything.

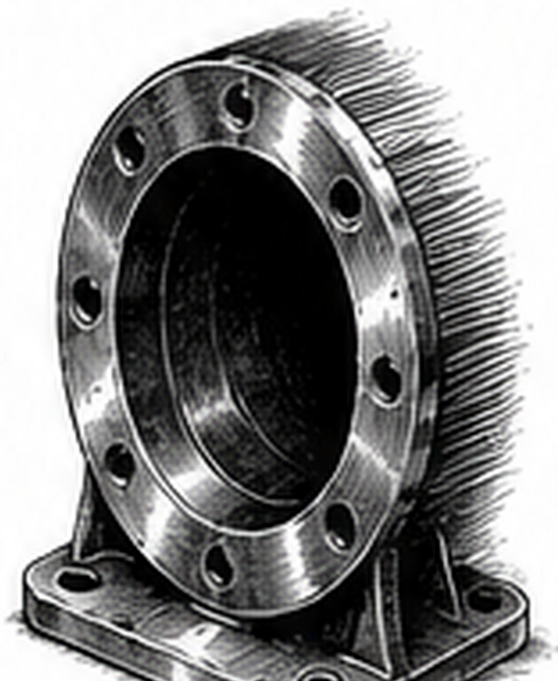
MANY CONSTRAINTS. NO OBVIOUS SOLUTION.

- It's 1970. An explosion cripples Apollo 13 on its way to the Moon.
- The crew is losing oxygen. Carbon dioxide is rising.
- The only working CO_2 filters don't fit the system.
- Engineers must fix it using only what's already onboard.
- No new parts. No second chances.

THE PROBLEM



Square cartridge.



Round socket.

The filter works.
It just doesn't fit.

EUREKA MOMENT

Constraints don't block the answer. They ***point*** to it.

THE CONSTRAINTS

- No spare parts
- No new materials
- Only what's on board
- Time is running out
- Lives depend on it

IDENTIFY CONSTRAINTS BEFORE SOLVING.

THE BUILD

Engineers on the ground and the crew in space improvised—using only the materials on board.

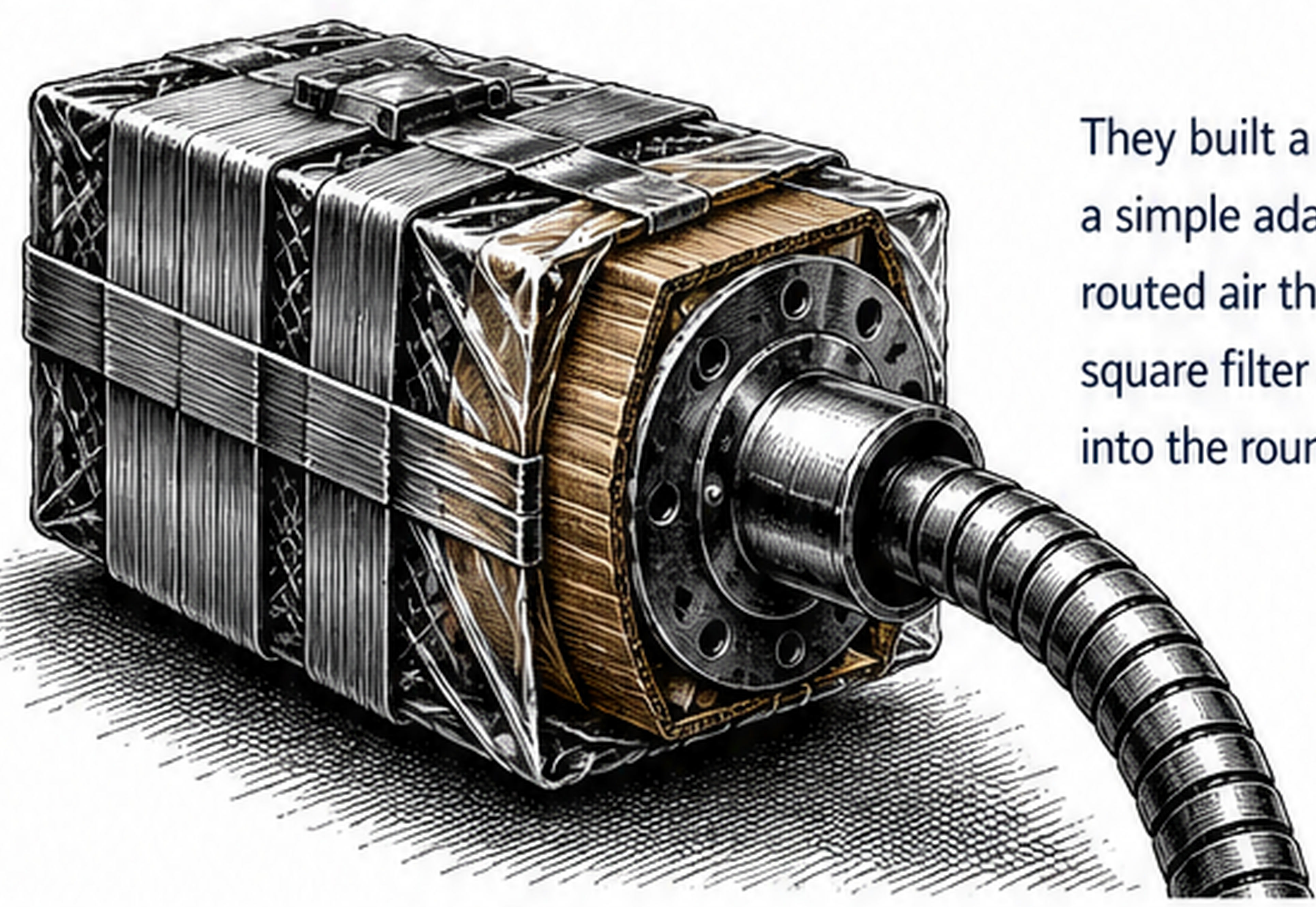
Duct tape

A sock

Cardboard

Plastic bags

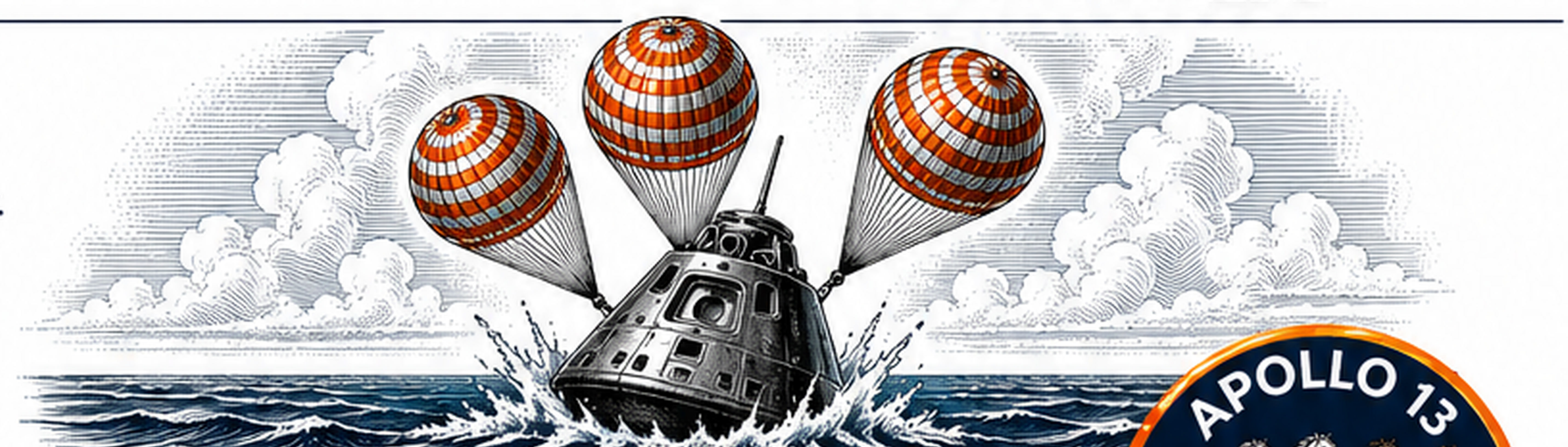
Hose



They built a "mailbox"—a simple adapter that routed air through the square filter and back into the round system.

THE RESULT

They rerouted the air.
CO_2 levels dropped.
The crew survived to return home.



GREAT PROBLEM SOLVERS DON'T IGNORE LIMITS—THEY USE THEM.